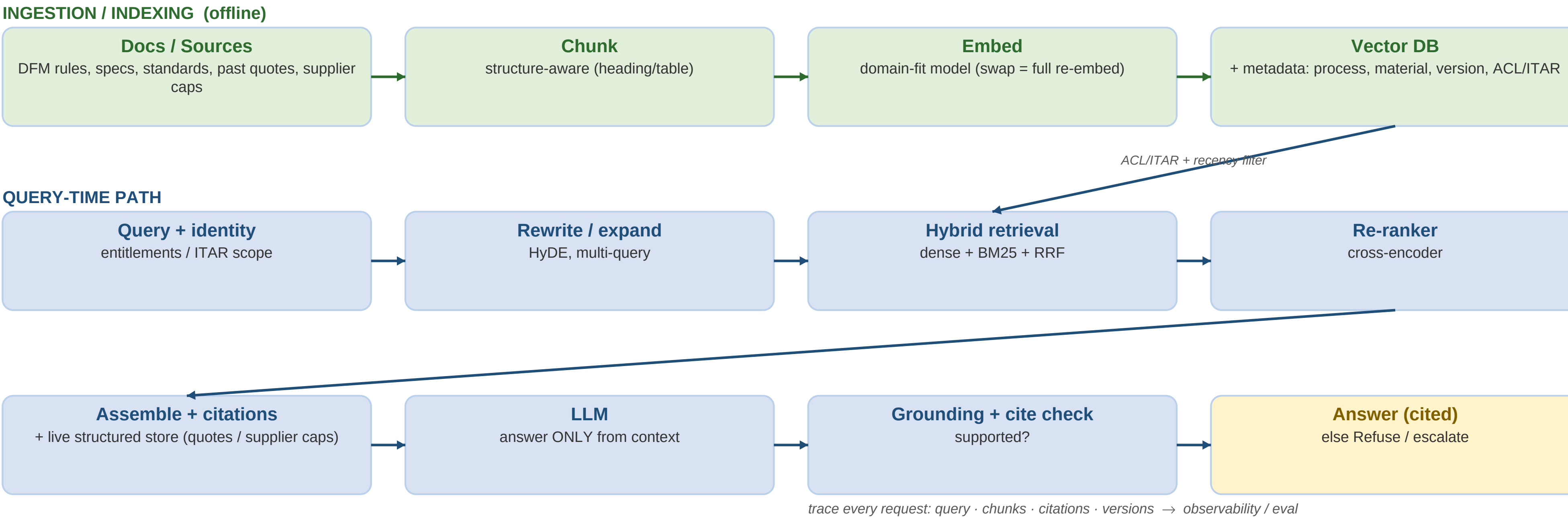
RAG System — One-Page Cheat Sheet

Xometry · Staff MLE (GenAI) · System Design — draw the flow, then grow it into each requirement while narrating tradeoffs



1 · Ingestion/Index — chunking (structure-aware)★ · size+overlap ~10-20% · embedding model · metadata (process/material/ACL) · HNSW vs IVF-PQ · parent→child

2 · Retrieval — top-k · dense vs BM25 vs HYBRID · RRF/weights · metadata+ACL filters · query rewrite / HyDE / multi-query

3 · Re-ranking — cross-encoder / ColBERT★ · shortlist-in (top 100) vs final-n kept → precision vs latency

4 · Generation — # chunks injected · context ordering (lost-in-middle) · “answer only from context” + cite + refuse · threshold · model+decoding

5 · Eval & Ops — recall@k · MRR/nDCG · faithfulness/groundedness · citation correctness · refusal rate · re-index cadence · caching

★ Two highest-leverage levers to reach for first: CHUNKING strategy + add a RE-RANKER.

GROW THE DIAGRAM IN 5 STAGES (narrate as you draw)

1. Baseline: query → embed → top-k → LLM answer.
2. “Wrong doc?” → hybrid (dense+BM25) + cross-encoder re-ranker + query rewrite.
3. “Stop hallucination?” → “answer only from context” + grounding/citation check (refuse if unsupported).
4. “ITAR / no cross-access?” → ACL/ITAR filter at retrieval + re-check before gen; self-host export-controlled data.
5. “Specs change weekly?” → incremental re-index + versioning + effective dates; embed swap = migration.

XOMETRY ONE-LINERS (say unprompted)

- ITAR → self-host + access-scope export-controlled data (drives model choice).
- Vector PDF vs raster scan → don’t OCR geometry already in the file.
- Name standards: ASME Y14.5 / ISO 1101.
- Frame errors in $: wrong tolerance = scrapped part; mis-retrieved spec = mis-quote.
- Confidence-gated human-in-the-loop → corrections become labels (flywheel).
- RAG over fine-tune: citations + freshness + auditability; FT only for style.